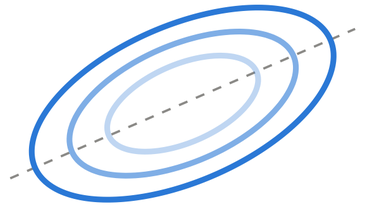
STAT3007 SEMINAR · GROUP G2

Scale or Shape?

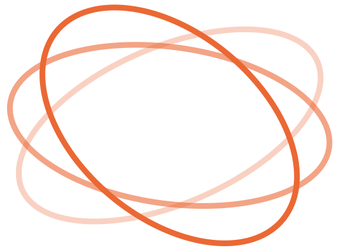
Decomposing the movement of the neural tangent kernel during grokking

Zain Al-Saffi · Jasper Chong · Aaron D'Mello
Zac Kienzle · Samuel Rieger

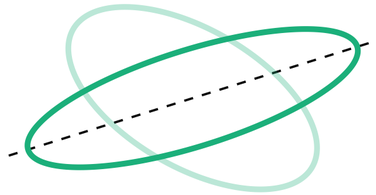
Semester 2, 2026



scale
the kernel grows



shape
the kernel changes shape



alignment
the kernel lines up with the labels

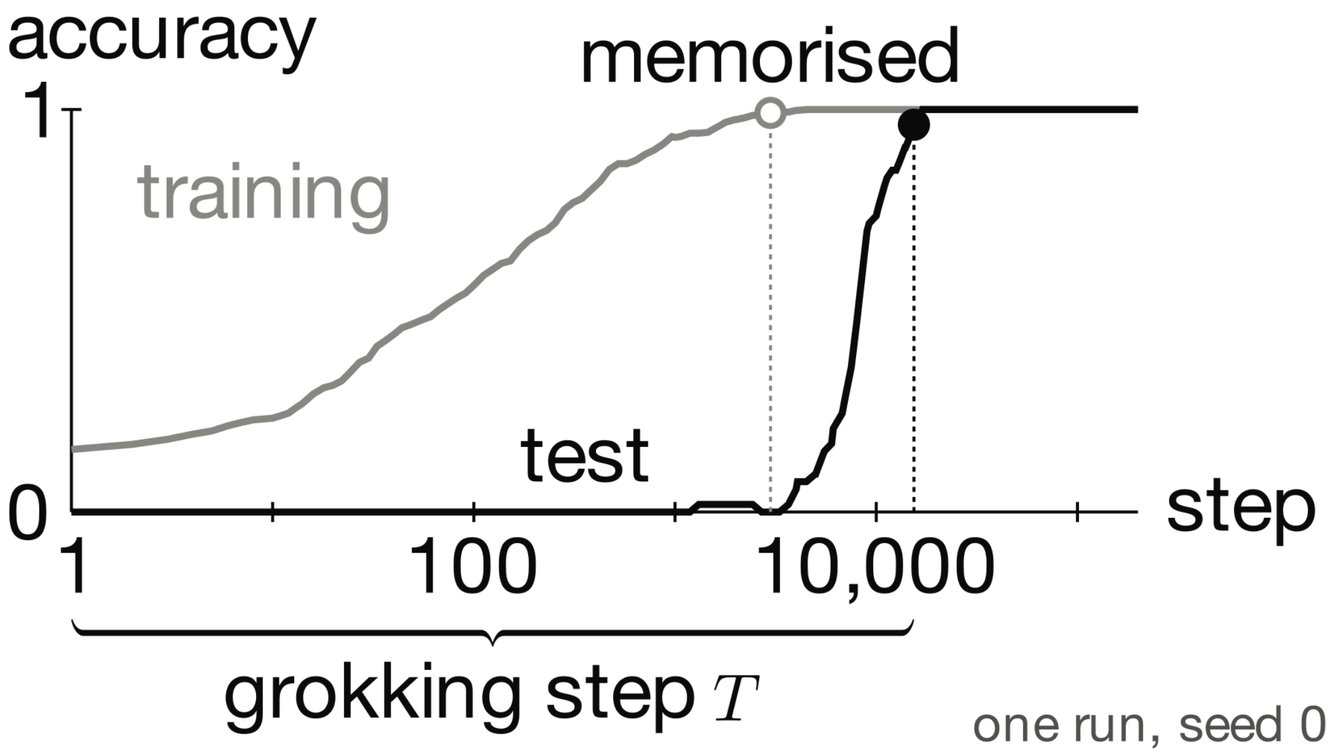
schematic

PART 1 OF 5

Grokking

A network fits its training data early. Why does it take so much longer to work on data it has not seen?

A network can fit its training data long before it generalises



Memorisation is the first step at which training accuracy reaches 0.99.

The grokking step \(T\) is the first step at which test accuracy reaches 0.95.

We repeated every analysis at test accuracy 0.80 and 0.90.

Grokking was first reported on small algorithmic datasets (Power et al. 2022).

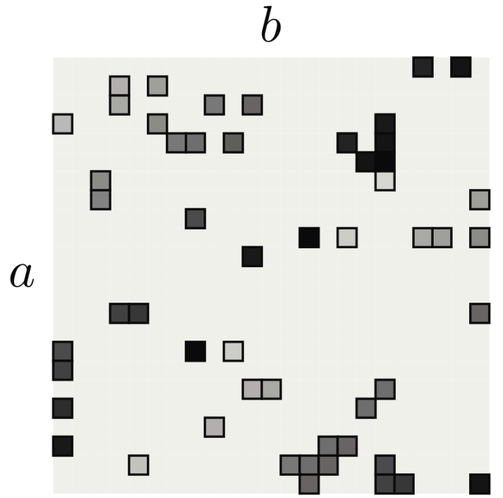
The task is modular addition, a standard test bed for grokking

- The input is a pair (a, b) with $a, b \in \{0, \dots, 22\}$. The target is $(a + b) \bmod 23$, so $(20 + 5) \bmod 23 = 2$.
- Each run trains on 476 of the 529 pairs and tests on the other 53.
- The network has one hidden layer of width N:

$$ f(x;\theta) = N^{-1/2}\, W^{(2)} \phi\!\left(D^{-1/2} W^{(1)} x\right) $$

$N^{-1/2}$ and $D^{-1/2}$ are the NTK parameterisation of Jacot et al. (2020).

Modular addition is a common test bed for grokking (Gromov 2023; Nanda et al. 2023; Mohamadi et al. 2024). Our setup is the baseline of Kumar et al. (2024, App. 8.3, arXiv v3).



476 training pairs

53 test pairs

Can the network fill these in?

PART 2 OF 5

The tangent kernel

What does it mean for a network to learn features, and how could we see it happen?

The tangent kernel shows which inputs the network treats alike

The tangent kernel is a snapshot of the features a network learns with.

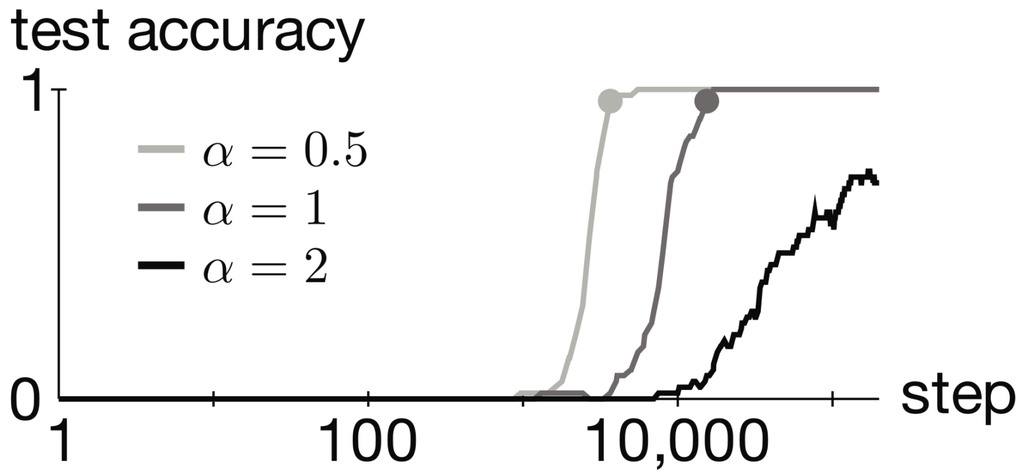
$$ K(x, x') = \big\langle \nabla_\theta f(x;\theta),\, \nabla_\theta f(x';\theta) \big\rangle $$

- A gradient step on x moves the output at x' in proportion to $K(x', x)$.
- Near its start the network is a linear model on the features $\nabla_\theta f$ (Chizat et al. 2020).
- If K stays fixed, training ends at kernel regression with K (Jacot et al. 2020).

Grokking may be the move from a fixed kernel to one that learns features

- **Lazy.** The kernel stays near its start, as in kernel regression.
- **Rich.** The kernel changes, and the network learns features.
- Grokking may be the move from lazy to rich (Kumar et al. 2024).
- A larger output scale \(\alpha\) keeps the network lazy for longer (Chizat et al. 2020).

\[\tilde f(x;\theta) = \alpha\,\big[f(x;\theta) - f(x;\theta_0)\big], \qquad \eta = \eta_0/\alpha^2 \]



A kernel can grow or change shape, and a distance cannot tell which

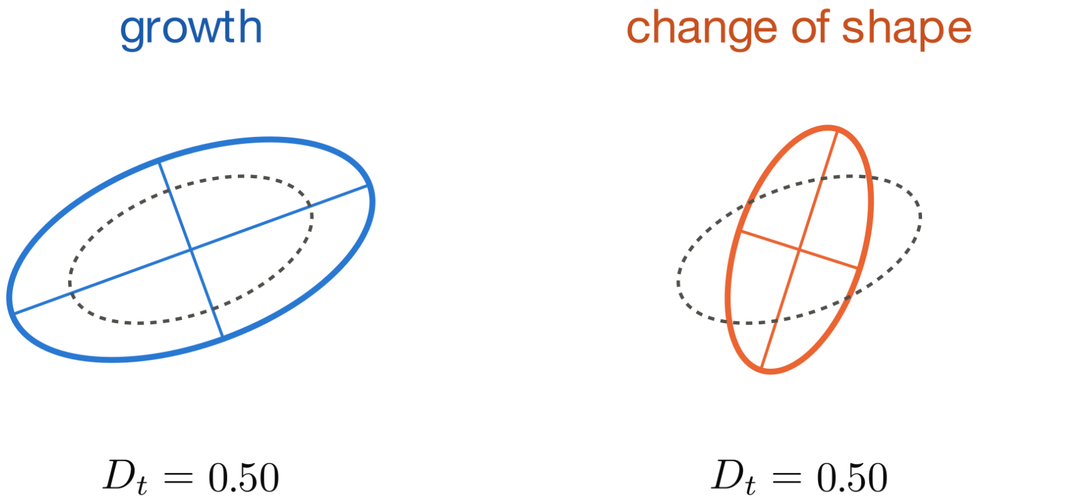
The usual measure of how far the kernel has moved is a distance (Geiger et al. 2020; Mohamadi et al. 2024).

$$ D_t = \frac{\Fnorm{K_t - K_0}}{\Fnorm{K_0}} $$

It splits exactly into a part for size and a part for shape.

$$ D_t^2 = \underbrace{(\Sc - 1)^2}_{\text{scale part}} + \underbrace{2\,\Sc\,\Rc}_{\text{shape part}} $$

> In our baseline, the scale part is 90% of D_t^2 at grokking.



Three numbers say how the kernel moved

SCALE

$$\Sc = \frac{\Fnorm{K_t}}{\Fnorm{K_0}}$$

The size of the kernel relative to the start, 1 when unchanged.

SHAPE

$$\Rc = 1 - \frac{\langle K_t, K_0 \rangle_F}{\Fnorm{K_t}\,\Fnorm{K_0}}$$

One minus the cosine between K_t and K_0 when treating them as long vectors. It is 0 when the kernel is only rescaled (Fort et al. 2020).

ALIGNMENT

$$\Ac = \frac{\langle CK_tC,\, CYY^\top C \rangle_F}{\Fnorm{CK_tC}\,\Fnorm{CYY^\top C}}$$

The cosine between the centred kernel and the label kernel $YY^\top$, which marks pairs with the same answer. It is high when the network treats inputs as alike exactly when they share an answer (Cortes et al. 2024).

Shape and alignment ignore rescaling.

K_t is the summed-logit kernel on the 53 test pairs of a run (Mohamadi et al. 2022).

PART 3 OF 5

What sets the clock?

Which kind of kernel movement sets the grokking time?

Every published clock is scale-like

A **clock** is the quantity that sets when a network groks.

WEIGHT NORM

The network groks once weight decay has shrunk the weight norm to a critical value (Khanh et al. 2026; Khanh 2026).

WEIGHT-DECAY RATE

The grokking time scales with \(1/(\eta\lambda)\) (Kim 2026; Pracher et al. 2026).

LAZY TO RICH

Grokking is the move to a kernel that adapts. Its evidence is a distance, which mostly records growth (Kumar et al. 2024).

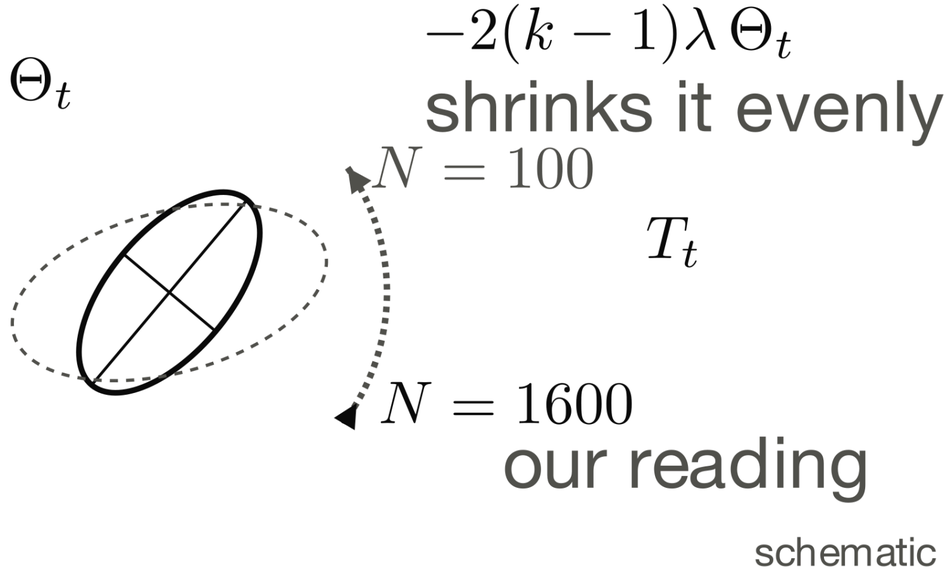
None of them measures the shape of the kernel. We ask which kind of kernel movement sets the time.

In theory, weight decay rescales and width slows the change of shape

$$\frac{dK_t(x,x')}{dt} = \underbrace{-2(k-1)\lambda\,K_t(x,x')}_{\text{scale part}} + \underbrace{T_t(x,x') + T_t(x',x)}_{\text{shape part}}$$

- Weight decay λ enters only the scale part, which shrinks the kernel evenly.
- Only T_t changes the shape, and at the start it is of order $1/N$.
- With $T \propto N^{a}(\eta\lambda)^{b}$, a scale clock gives $b = -1$ (Kim 2026). A shape clock gives $a > 0$ and $b \approx 0$ (our prediction).

Eqs. S5 and S6 of Lewkowycz and Gur-Ari (2021, arXiv v2), for gradient flow on a network whose output is homogeneous of degree k in its weights. For our network it is a guide.



Our hypothesis: the shape of the kernel sets the grokking time

Hypothesis. A network groks once its kernel has changed shape enough, so the shape statistic \(R_t\) sets the clock.

Test	Shape clock (our hypothesis)	Weight-decay clock	Scale clock
Tier 1. Is a statistic the same at every grokking step?	Yes, \(R_t\)	Not stated	Yes, \(S_t\) (our prediction)
Tier 2. How does \(T\) scale?	Wider networks grok later, and weight decay matters little (our prediction)	\(T \propto 1/(\eta\lambda)\), and width does not matter (our prediction)	Not stated
Tier 3. What predicts \(T\) early?	\(R_t\)	\(\eta\lambda\), which a real network does not report	\(S_t\)

“Our prediction” marks a prediction that the source does not state, which we derived from it.

We trained 720 networks across laziness, width and weight decay

Setting	Values	What it tests
Output scale \(\alpha\)	0.5, 1, 2	How lazy training is
Width \(N\)	50 to 1600, six values	How fast the shape can change
Weight decay \(\eta\lambda\)	0, and seven values from \(3\times10^{-6}\) to \(3\times10^{-3}\)	The weight-decay clock

Each of the 144 cells has 5 seeds, and each run has 200,000 steps.

A run that has not grokked by then is kept as **censored**, with \(T > 200{,}000\). The fits use it through the likelihood (Kalbfleisch and Prentice 2002).

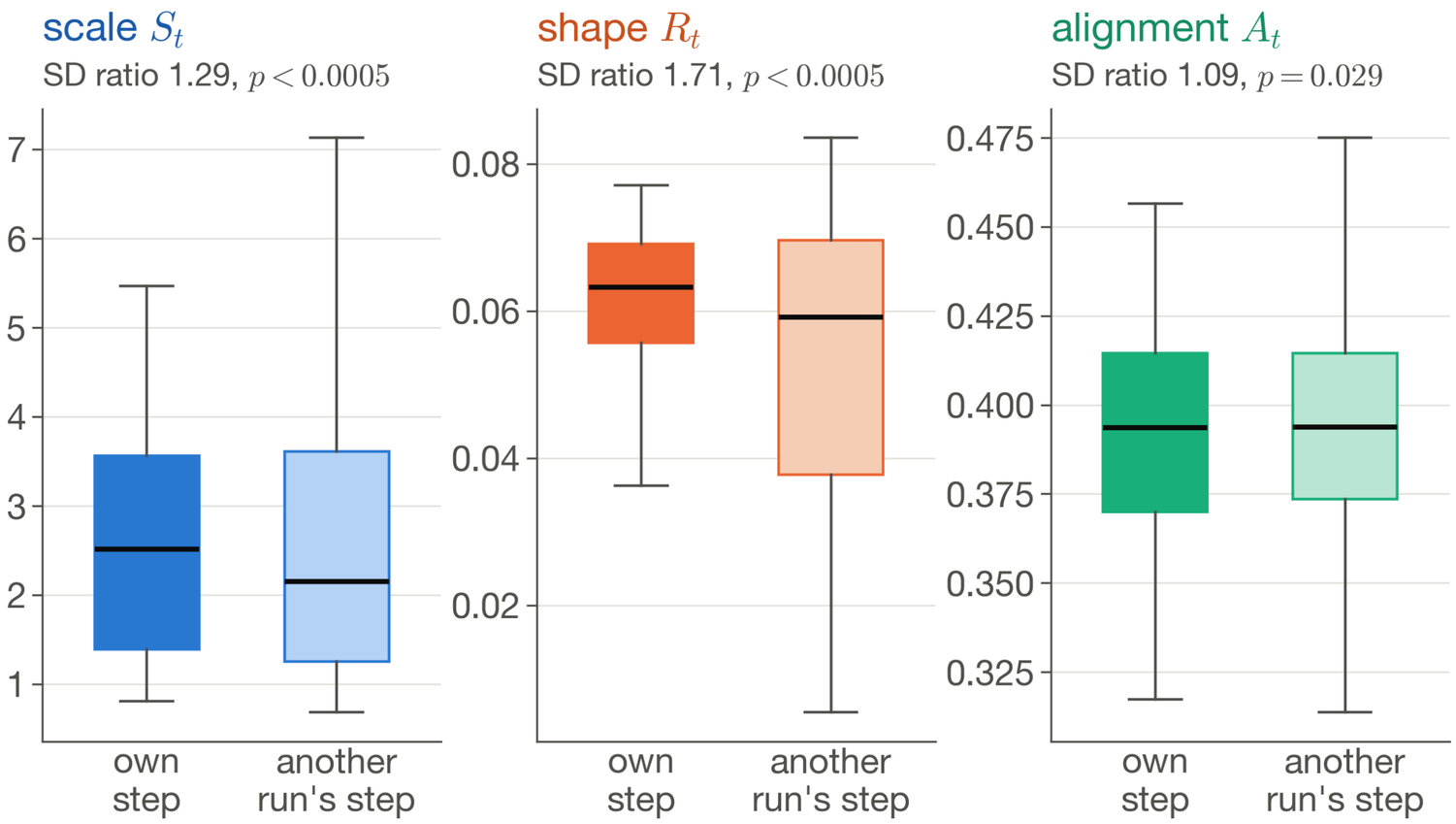
Our baseline grokking steps are close to those of Kumar et al. (2024, fig. 2(c), arXiv v3).

PART 4 OF 5

Results

Which clock does the grokking time follow?

Tier 1: at a fixed width, shape is the statistic most tied to grokking



If grokking has a boiling point, a statistic reads the same in every run at the moment it groks.

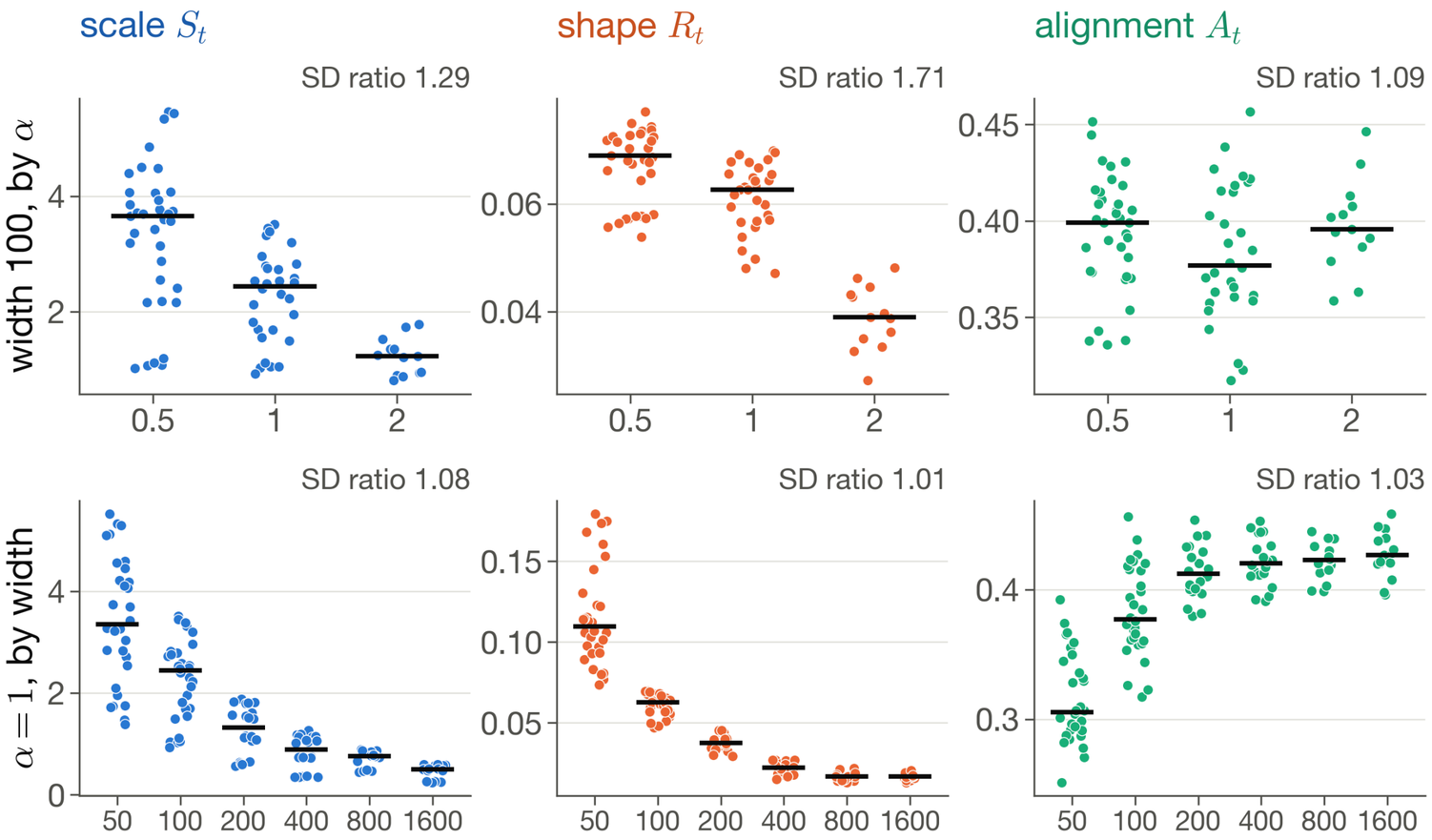
We also read each run at another run's grokking step. The **SD ratio** says how much more the readings spread there.

The 78 runs of width 100 that grok. Boxes hold the middle half of the runs, and whiskers reach 1.5 box heights.

FOR THE CLOCK

Shape bunches up the most at the true grokking step, and then scale. Alignment hardly changes.

No statistic is a fixed threshold, so shape is not the whole clock



- At width 100 (top), shape bunches up at grokking.
- Across widths (bottom), no statistic keeps one value at grokking, and no \(p\) is below 0.05.
- Wider networks grok with less change of shape, yet later.

FOR THE CLOCK

Shape tells us the most. Shape alone does not set the grokking time.

Tier 2: the grokking time follows both width and weight decay

$$\log T = c + a\log N + b\log(\eta\lambda) + \gamma\log N\log(\eta\lambda) + \sigma\varepsilon$$

Each clock predicts $T \propto N^a(\eta\lambda)^b$, a straight line in logarithms. γ lets the settings interact, and its interval **includes 0**.

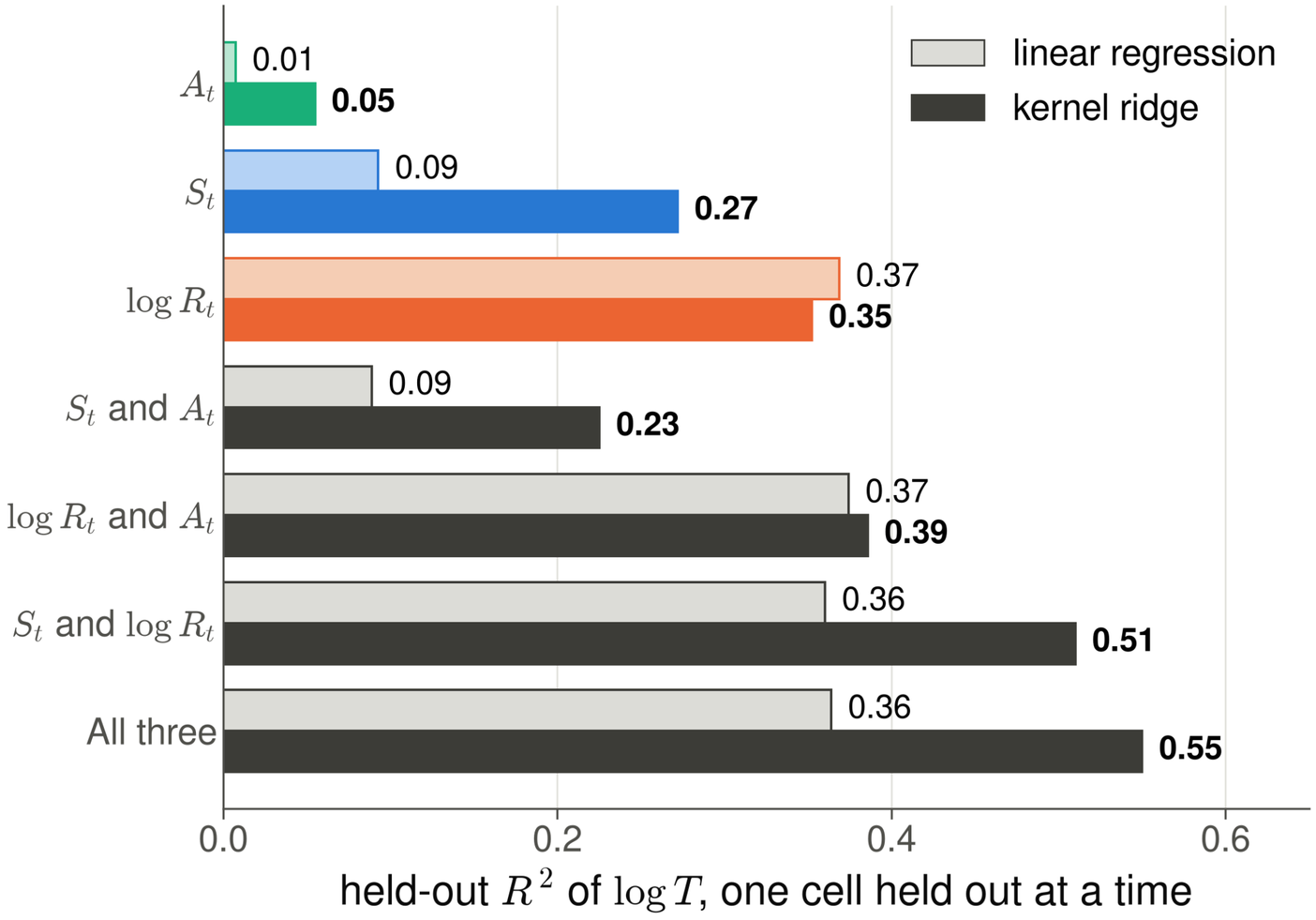
	a (width)	b (weight decay)	γ	σ
Shape clock predicts	> 0	≈ 0		
Weight-decay clock predicts	≈ 0	-1		
Our fit	**$+0.24$**	**-0.63**	-0.16	0.34
95% interval	$+0.08$, $+0.40$	-0.92, -0.46	-0.37, $+0.05$	

FOR THE CLOCK

The fit takes $a > 0$ from the shape clock and $b < 0$ from the weight-decay clock. The clock is mixed.

A censored log-normal fit on the 90 runs at $\alpha = 1$ with $10^{-5} \le \eta\lambda \le 10^{-4}$, where every seed groks.

Tier 3: shape is the best single predictor, and scale adds what shape lacks



We read the statistics at step 2,430, before any run groks, and predict \(\log T\) for settings the model has not seen. \(R^2\) is the share of the spread in \(\log T\) that the prediction explains.

In a straight line, scale adds nothing to shape. Kernel ridge can bend scale and combine it with shape.

FOR THE CLOCK

Shape alone predicts best. Scale helps once the model can bend it, and then \(R^2\) rises from 0.35 to 0.51. The clock is mixed.

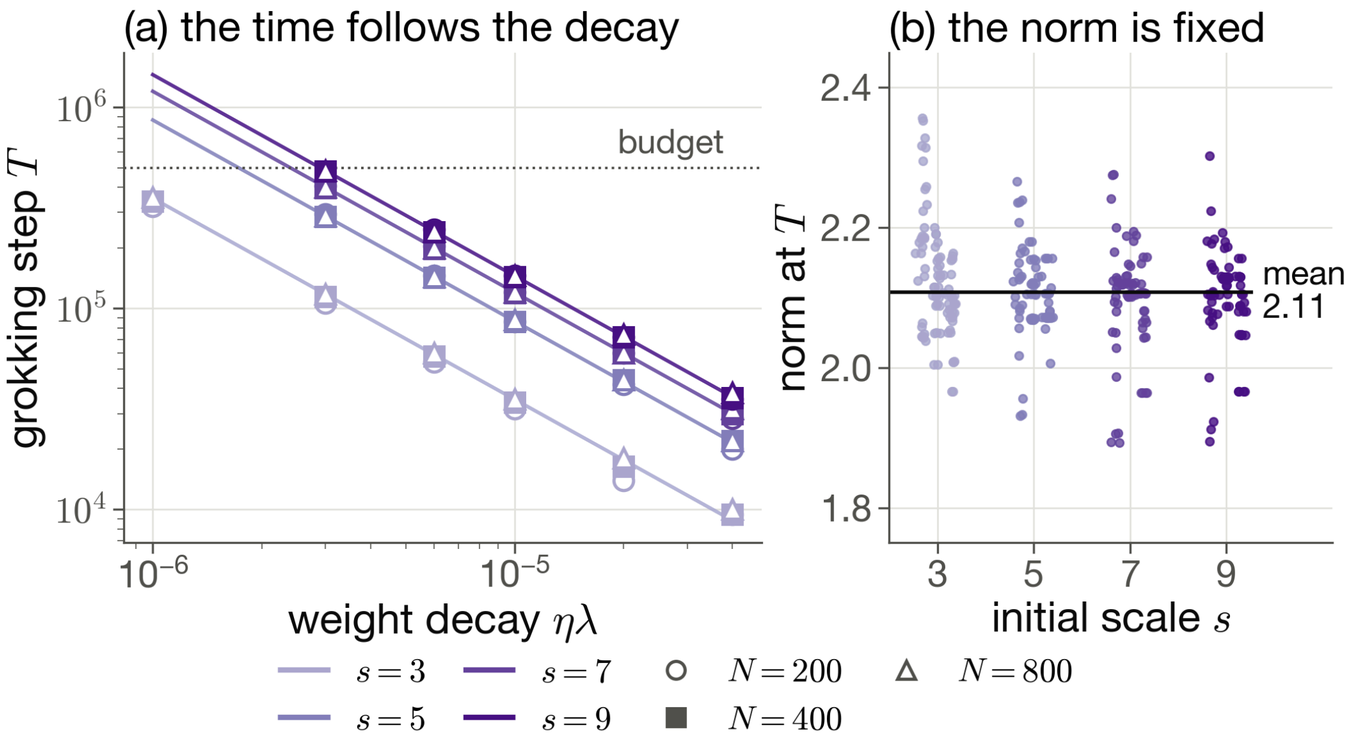
A straight line with scale squared reaches 0.50, and with the products of the statistics 0.54. We use \(\log R_t\), since with \(R_t\) as it is all three reach only 0.47.

Tier 4: on MNIST, weight decay alone sets the time

We repeated Tiers 1 and 2 on MNIST in the setup of Liu et al. (2023).

Setting	Value
Data	1,000 training images
Network	three layers, rectifier, no biases, squared error
Initial scale \(s\)	3, 5, 7, 9
Width \(N\)	200, 400, 800 in the fits
Weight decay	seven values, 5 seeds each
Budget and level	500,000 steps, test accuracy 0.80

Large initial weights must shrink to a fixed norm, so \(T \propto 1/(\eta\lambda)\) and width drops out.



FOR THE CLOCK

MNIST does not force the kernel to change shape, so only the scale route acts. Modular addition does (Mohamadi et al. 2024), so both act (our reading).

PART 5 OF 5

What we found

So what sets the grokking time?

Limitations

- One task, one architecture and the NTK parameterisation.
- No seed groks in 70 of the 144 cells, and the exponents depend on which cells are fitted.
- The settings predict better than the kernel statistics.
- MNIST changes the task, the network and how the delay is made at once.

Grokking runs on a mixed clock, and shape is its best single reading

GROWTH

The usual kernel distance mostly records growth, so split it into scale and shape.

SHAPE

Shape is the most informative statistic. It is not enough alone and not a fixed threshold.

A MIXED CLOCK

No single account holds, ours included. Width, through shape, and weight decay, through scale, both set the time.

On MNIST weight decay alone sets the time. That shape is needed only where a fixed kernel cannot solve the task is our reading (Mohamadi et al. 2024).

Thank you

Questions?

The code and the data extract behind every number are in the repository, in `report_v2/code`.

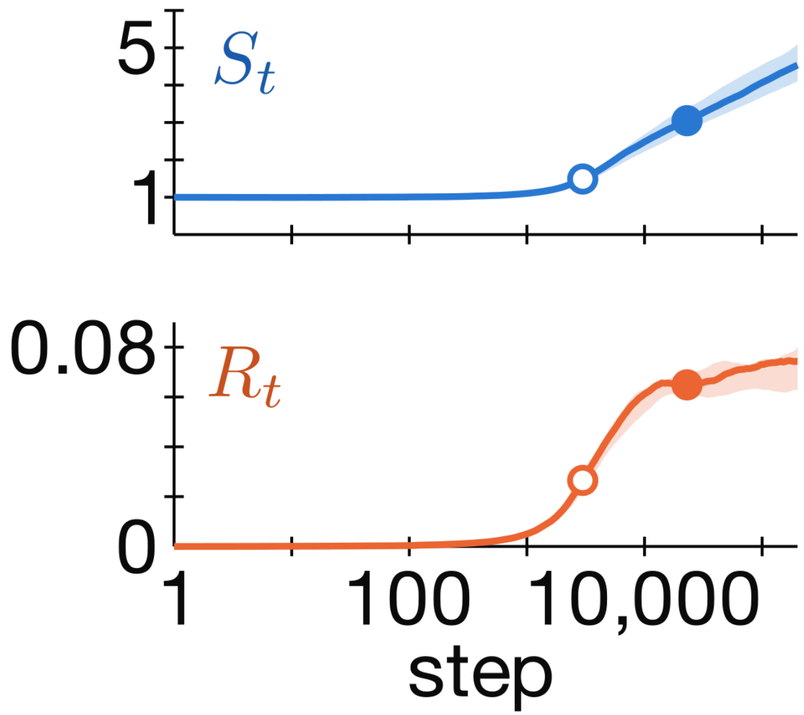
Backup

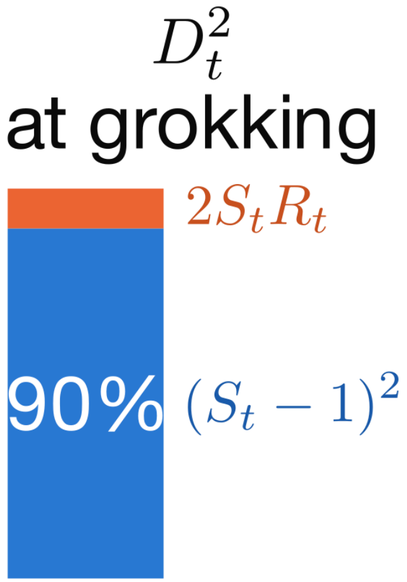
These slides are for questions.

How we used AI tools

TODO: agree this AI statement as a group We used an AI assistant (Claude, by Anthropic) to help draft code, the report and this deck. We checked every result and every reference ourselves.

Baseline: the network memorises early, groks late, and its kernel mostly grows

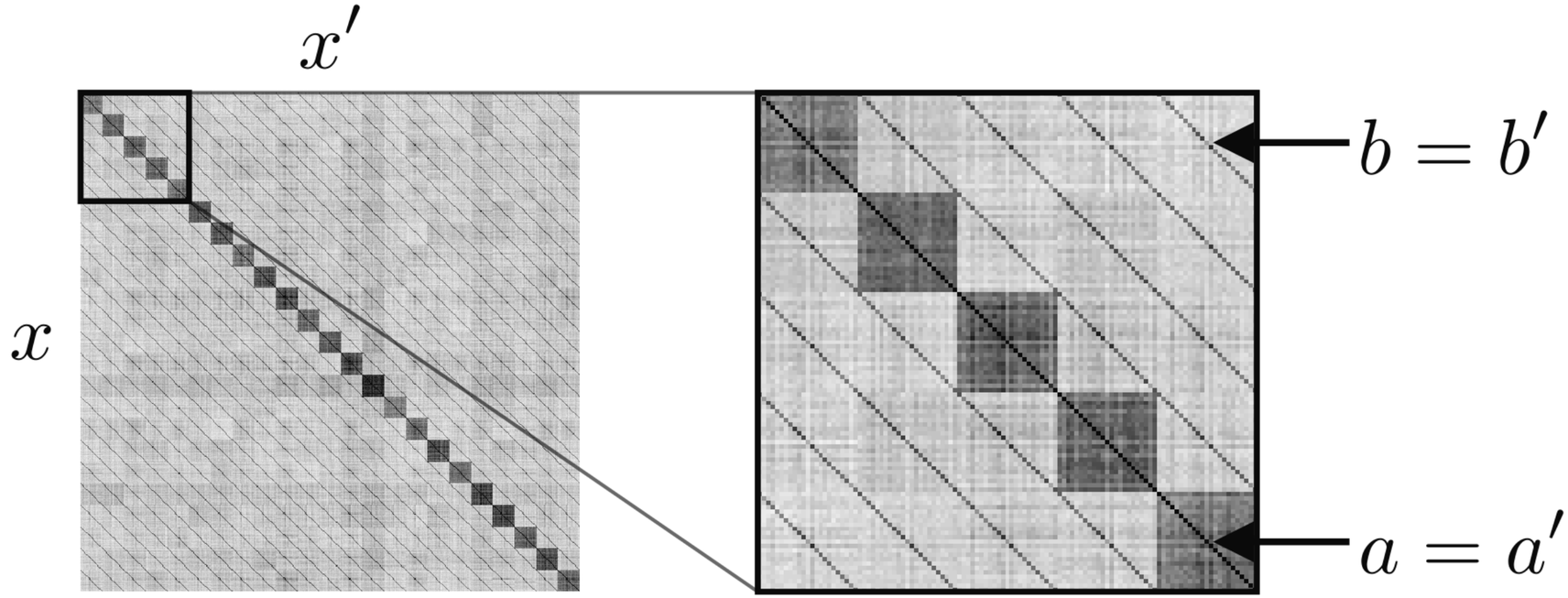




- At grokking, the scale part is 90% of \(D_t^2\).
- By the last step the kernel is 4.5 times its starting size, while its shape statistic is only 0.074.
- At \(\alpha = 2\) no seed groks within 200,000 steps.

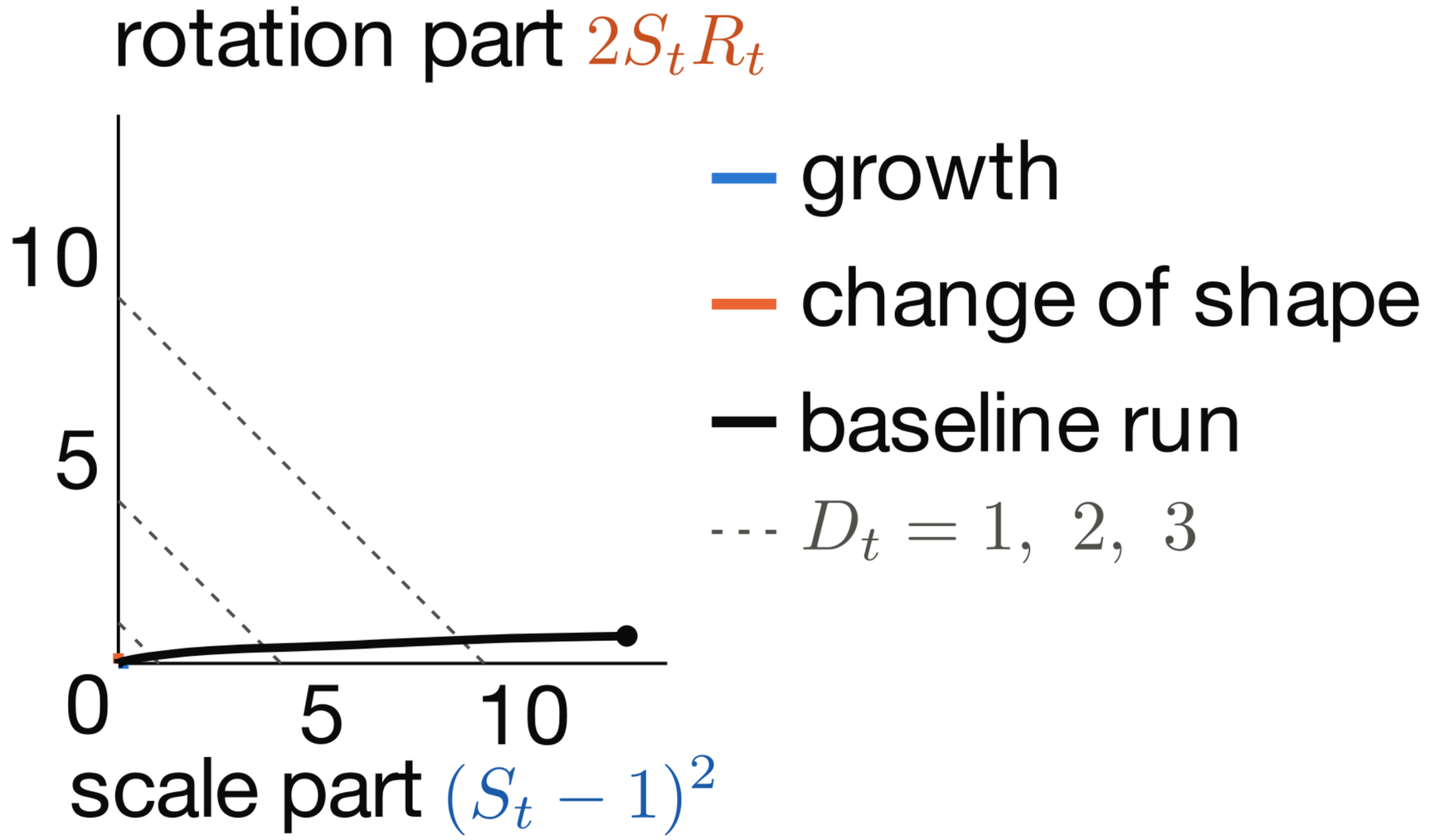
Our grokking steps, and their order in \(\alpha\), are close to those of Kumar et al. (2024, fig. 2(c), arXiv v3).

The tangent kernel of one network at the start

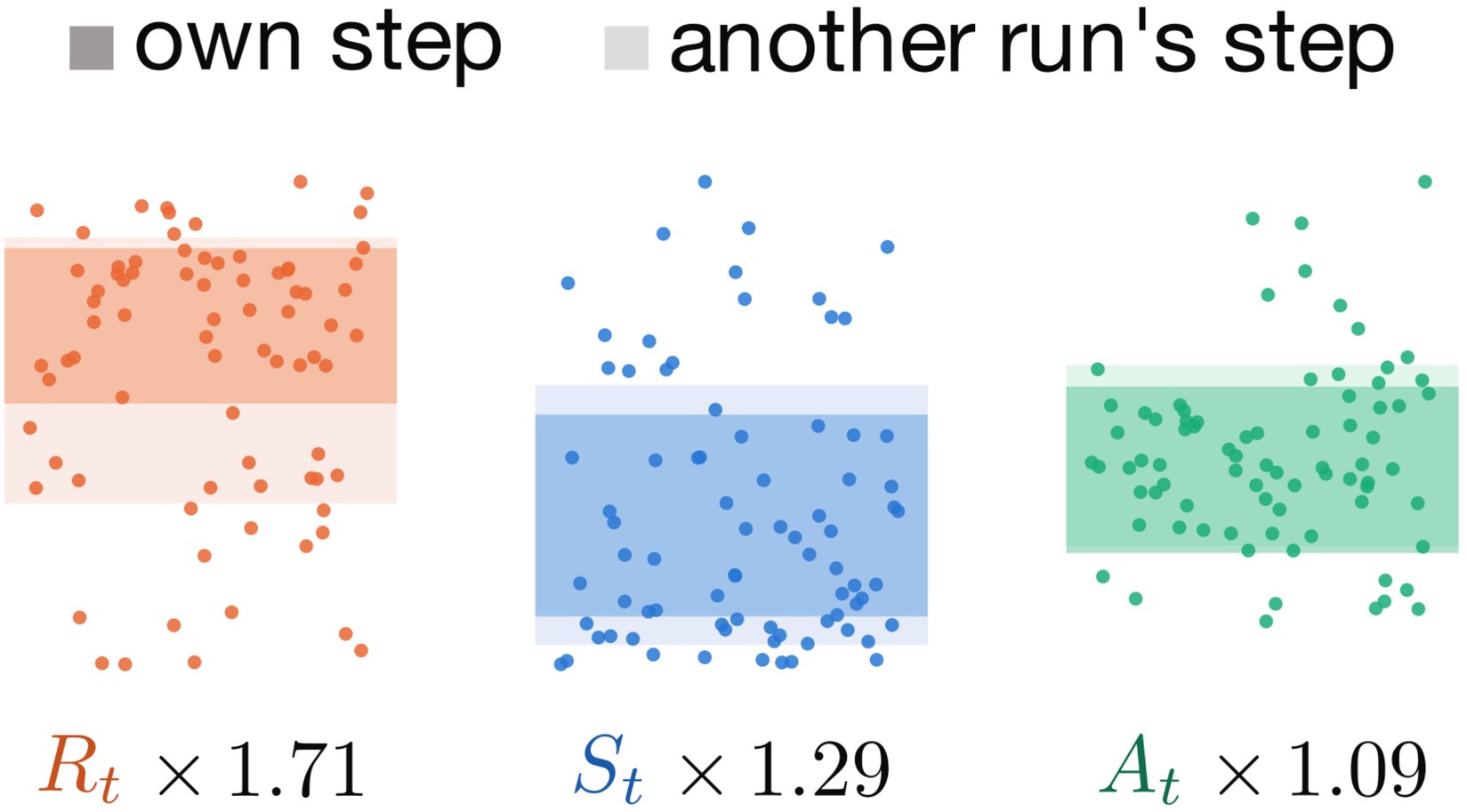


measured: one network at step 0

The split of the distance on a plane

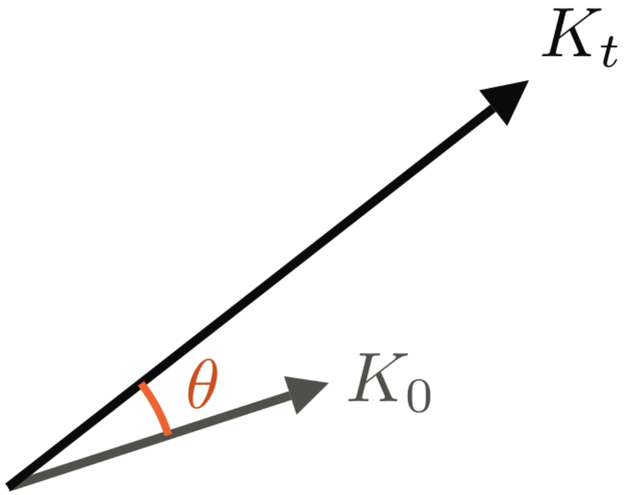


The placebo test, step by step



Seen as arrows, the three numbers are a length and two angles

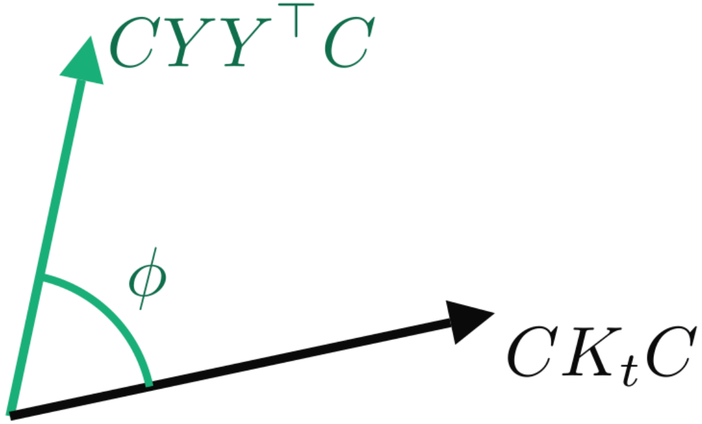
the kernel



$$S_t = \frac{\|K_t\|_F}{\|K_0\|_F} = 1.96$$

$$R_t = 1 - \cos\theta = 0.06$$

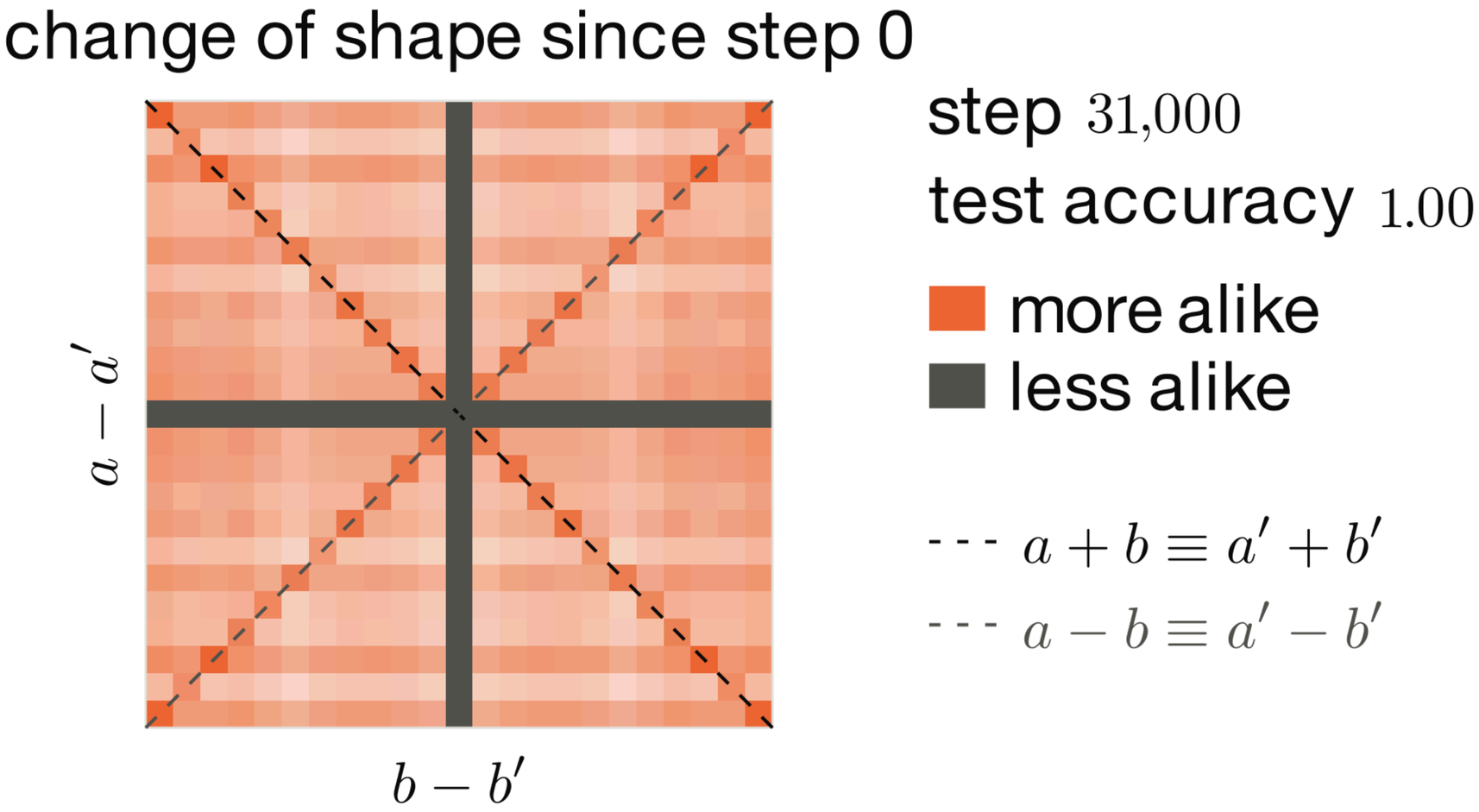
after centring



$$A_t = \cos\phi = 0.41$$

schematic

On this task, the change of shape has a visible pattern



Three tests, one for each row of the table

TIER 1

Does grokking happen at a fixed value of a statistic? A placebo test reads each run at another run’s grokking step.

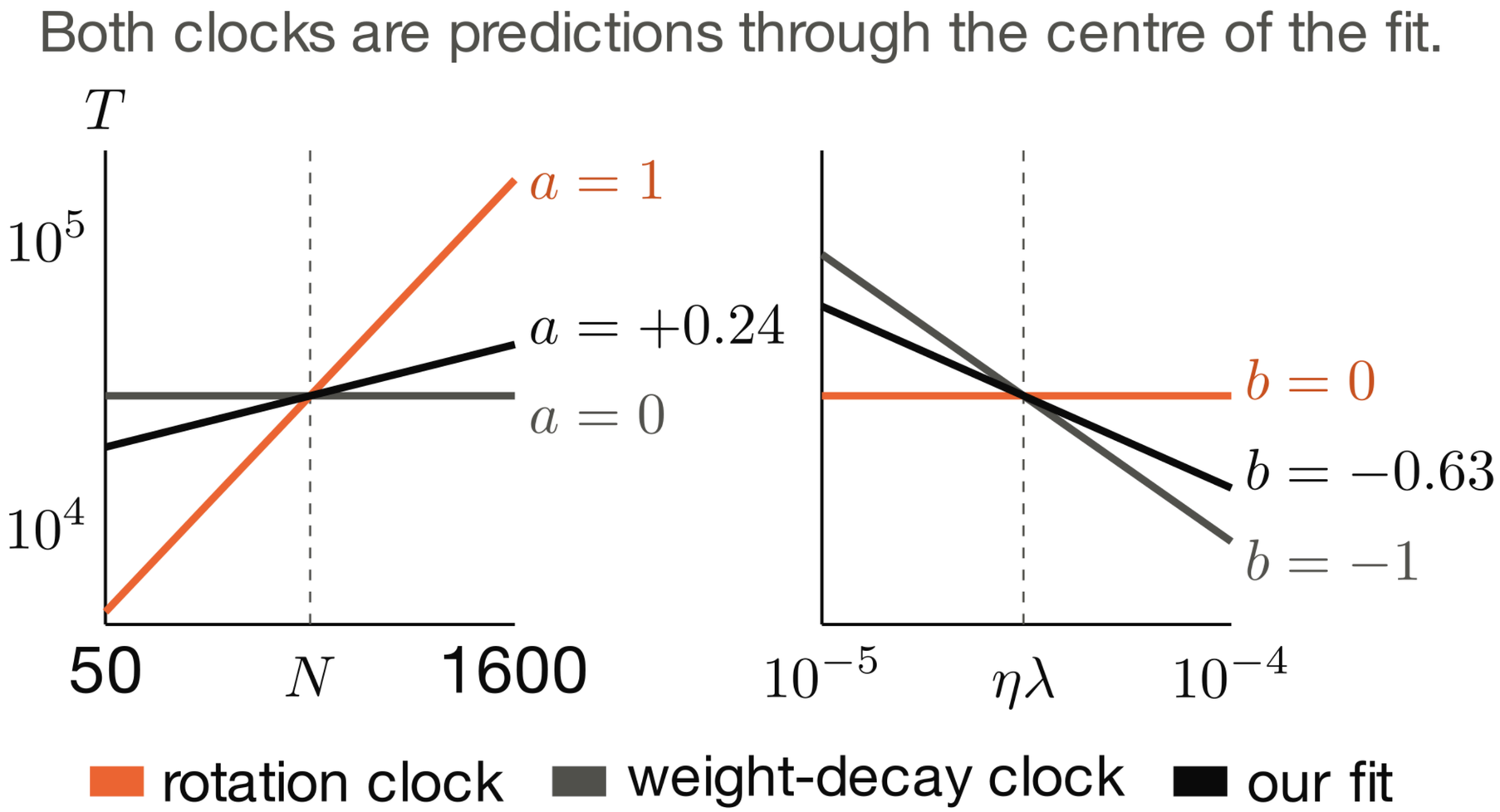
TIER 2

How does \(T\) scale with width and weight decay? We fit a censored log-normal model of \(\log T\).

TIER 3

Which statistic, read before any run groks, predicts \(T\)? Models are scored on cells and on values of \(\alpha\) they have not seen.

What the two clocks would look like

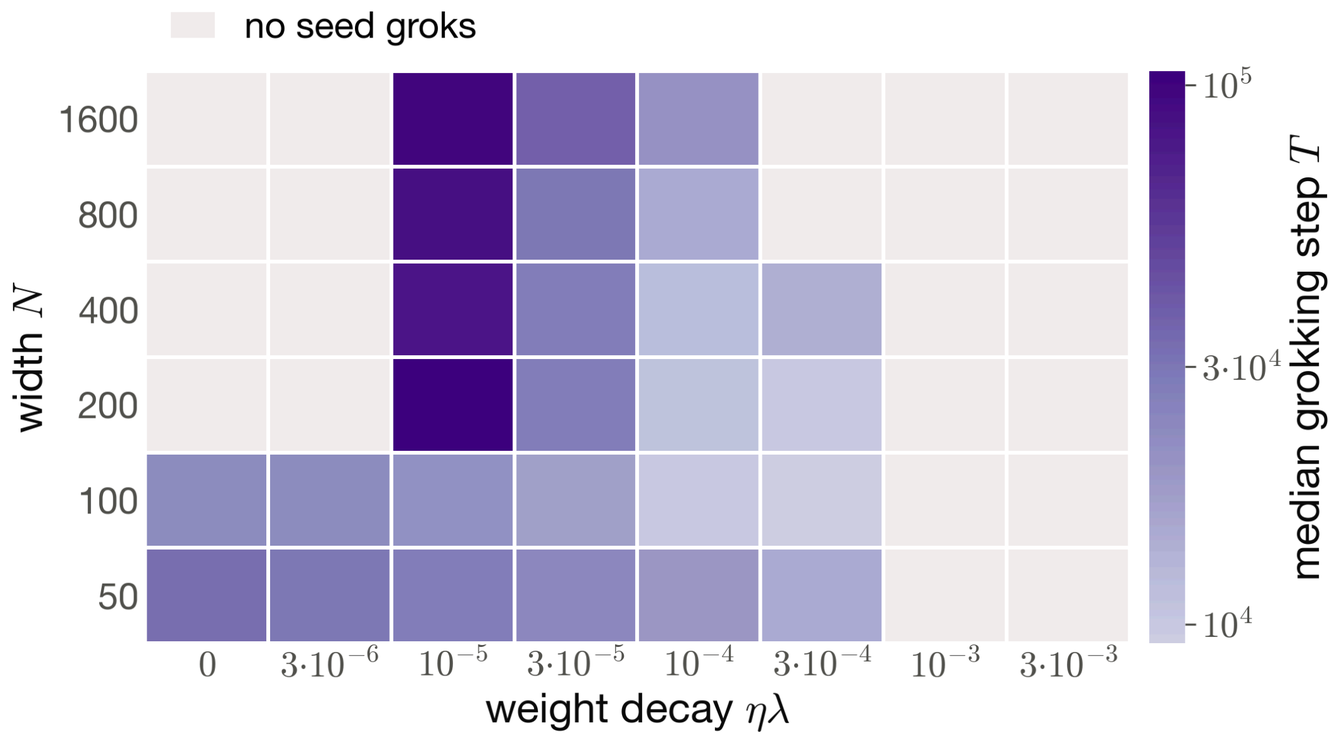


Tier 2: the grokking time at each width



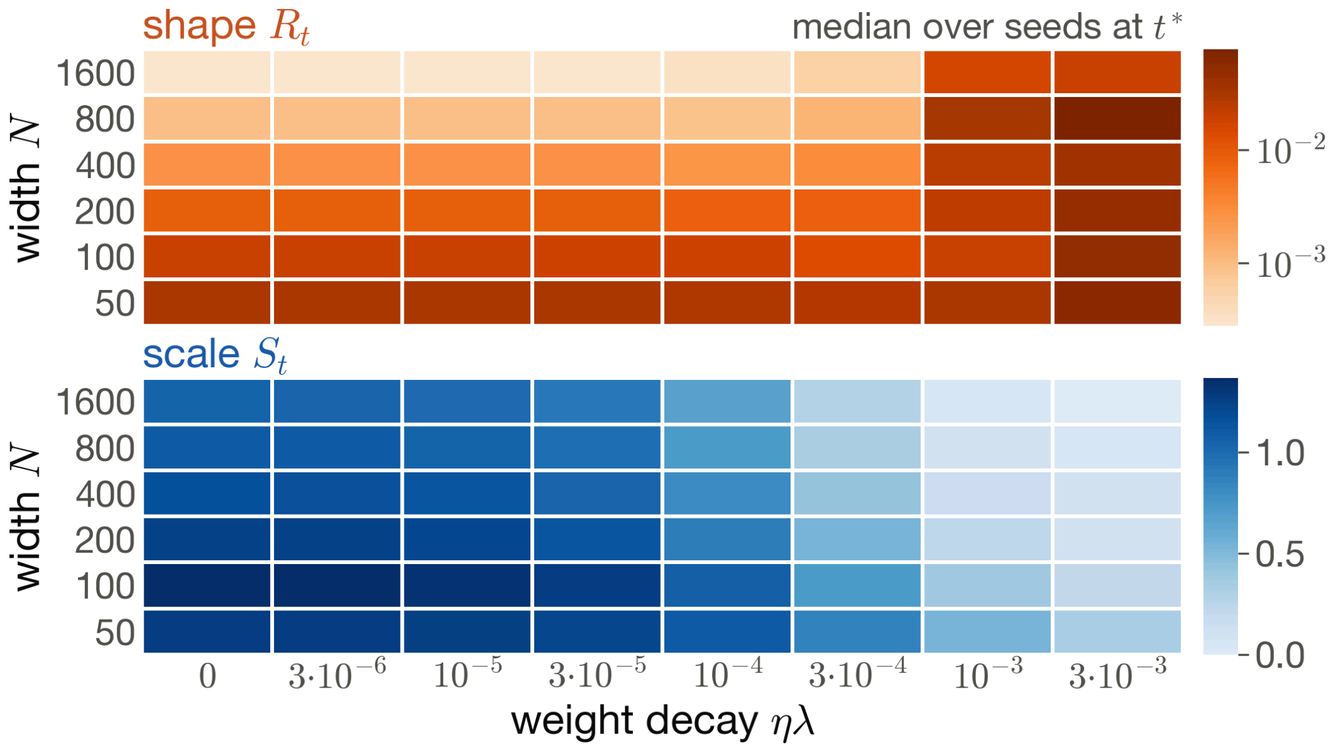
- Each line is one width, and each point is the median grokking step of a cell.
- Open circles at the top did not grok within 200,000 steps.
- The grey lines show the slope each clock predicts. In the shaded fit range the lines follow neither slope consistently.

A wider network groks later and stops grokking at a smaller weight decay



- Without weight decay, only \(N = 50\) and \(N = 100\) grok within the budget.
- Every seed groks up to \(\eta\lambda\) = \(3\times10^{-4}\) for \(N \le 400\), and only up to \(10^{-4}\) for \(N = 800\) and \(1600\).
- The weight decay that speeds grokking depends on the width.

Width sets the shape change, and at a fixed width weight decay shrinks the scale

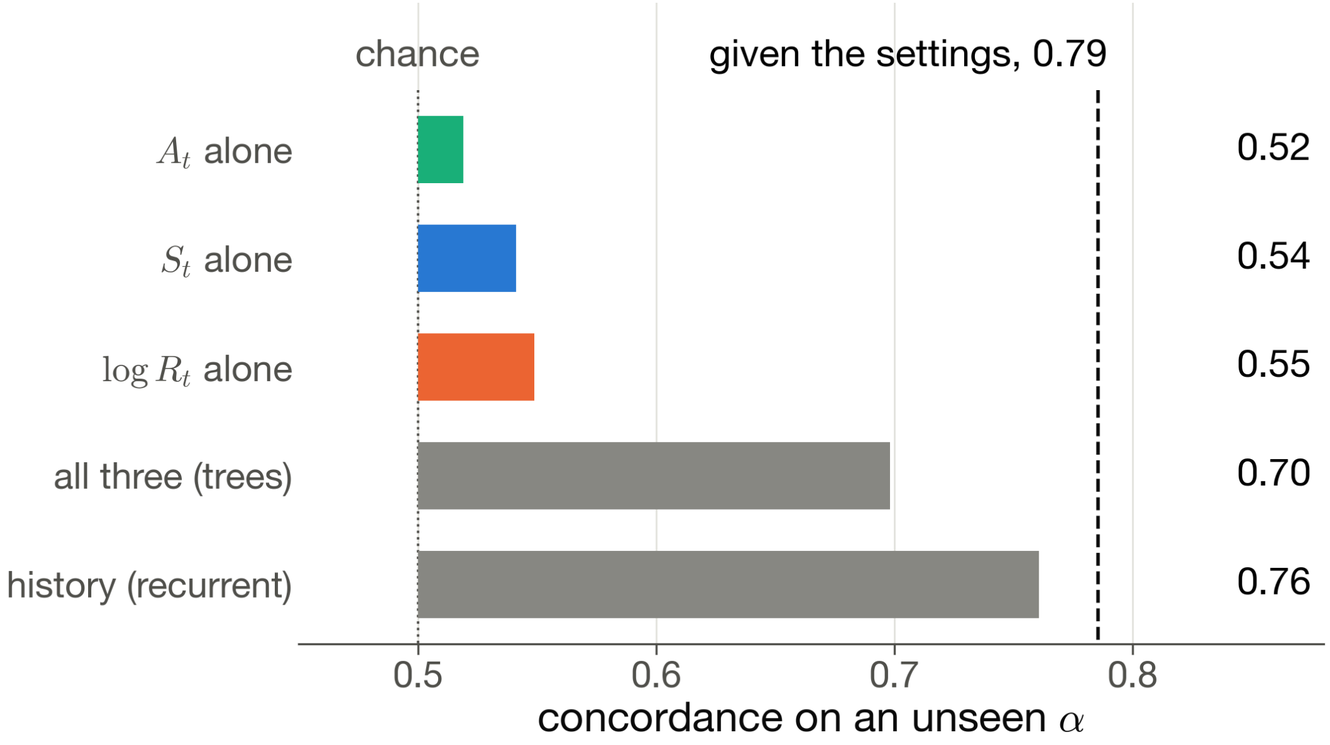


Both maps are read at \(t^* = 2{,}430\), before any run groks.

- Up to \(\eta\lambda = 3\times10^{-4}\), the shape change varies with the width and hardly with the weight decay.
- The scale falls with the weight decay at every width.

Among runs that grok, width explains 0.66 of the spread of \(\log R_t\), and weight decay 0.18. The spread of \(S_t\) is shared: \(\alpha\) explains 0.28, width 0.22 and weight decay 0.26.

On an unseen \(\alpha\), the statistics work best together, and so do width and weight decay



- Each statistic alone ranks runs little better than chance: at most 0.55, where 0.5 is chance.
- Boosted trees on all three reach 0.70. A recurrent network that reads their history up to \(t^*\) reaches 0.76.
- As a reference, trees given width and weight decay reach 0.79, above every kernel model. Given width alone they reach 0.67, and given weight decay alone 0.72.

> Each setting alone ranks runs worse than the two together. This supports the mixed clock of Tier 2.

The score is the concordance, the share of pairs of runs put in the right order. Bars are means over the three held-out values of \(\alpha\).

The identity behind the split

$$\Fnorm{K_t - K_0}^2 = \Fnorm{K_t}^2 - 2\langle K_t, K_0\rangle_F + \Fnorm{K_0}^2$$

Divide by $\Fnorm{K_0}^2$ and use $\langle K_t, K_0\rangle_F = (1 - R_t)\,\Fnorm{K_t}\,\Fnorm{K_0}$:

$$D_t^2 = S_t^2 - 2S_t(1 - R_t) + 1 = (S_t - 1)^2 + 2S_tR_t .$$

By the Cauchy-Schwarz inequality, $R_t = 0$ exactly when $K_t = cK_0$ for some $c > 0$, and then $D_t = |S_t - 1|$.

The placebo test in numbers

Runs	Statistic	SD at grokking	SD at placebo	SD ratio	p
$N = 100$ (78 runs)	S_t	1.2	1.6	1.29	<0.0005
	R_t	0.012	0.02	1.71	<0.0005
	A_t	0.031	0.034	1.09	0.029
$\alpha = 1$ (130 runs)	S_t	1.3	1.4	1.08	0.258
	R_t	0.041	0.041	1.01	0.423
	A_t	0.05	0.052	1.03	0.079

The placebo SD is the median over 2,000 permutations with no fixed points. p is the share of permutations with an SD no larger than the true one.

The Tier 2 fit and its checks

Fit at \(\alpha = 1\)	**\(a\)**	**\(b\)**	**\(\gamma\)**
Main range, level 0.95	\(+0.24\) [\(+0.08\), \(+0.40\)]	\(-0.63\) [\(-0.92\), \(-0.46\)]	\(-0.16\) [\(-0.37\), \(+0.05\)]
Adding \(\eta\lambda = 3\times10^{-4}\)	\(+0.36\) [\(+0.03\), \(+0.62\)]	\(-0.21\) [\(-0.60\), \(+0.14\)]	\(+0.14\) [\(-0.24\), \(+0.48\)]
Level 0.80	\(+0.30\) [\(+0.17\), \(+0.45\)]	\(-0.55\) [\(-0.74\), \(-0.39\)]	\(-0.17\) [\(-0.36\), \(-0.02\)]
Level 0.90	\(+0.27\) [\(+0.13\), \(+0.41\)]	\(-0.60\) [\(-0.84\), \(-0.42\)]	\(-0.16\) [\(-0.37\), \(+0.03\)]
Memorisation step	\(-1.53\) [\(-1.80\), \(-1.32\)]	\(+0.07\) [\(-0.13\), \(+0.31\)]	\(+0.05\) [\(-0.23\), \(+0.29\)]

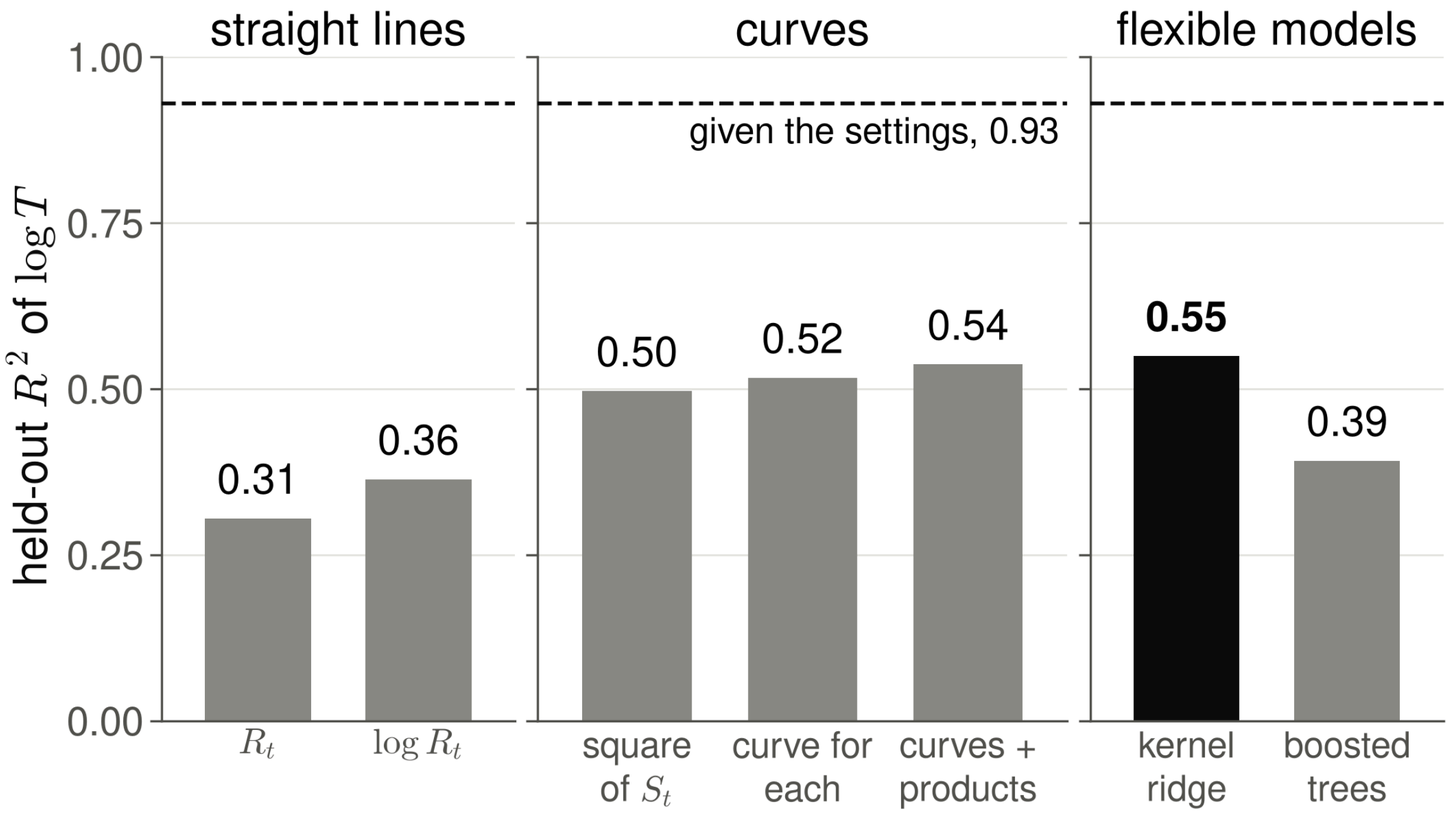
Brackets give 95% intervals from a bootstrap that resamples cells, then seeds. The interval of \(\gamma\) includes zero in the main range, so we do not claim its sign. Wider networks memorise sooner, so the width effect on grokking is not a delay in fitting the training data.

Tier 3 on held-out cells in numbers

Statistics at \(t^*\)	Held-out \(R^2\)	Concordance
\(A_t\)	0.05	0.55
\(S_t\)	0.27	0.67
\(\log R_t\)	0.35	0.70
\(S_t\) and \(A_t\)	0.23	0.65
\(\log R_t\) and \(A_t\)	0.39	0.72
\(S_t\) and \(\log R_t\)	0.51	0.76
All three	0.55	0.78

The share of the spread that lies between cells is 0.96 for \(\log T\), 0.99 for \(S_t\), 0.99 for \(\log R_t\) and 0.56 for \(A_t\).

Tier 3: a curve beats a straight line, and kernel ridge fits best



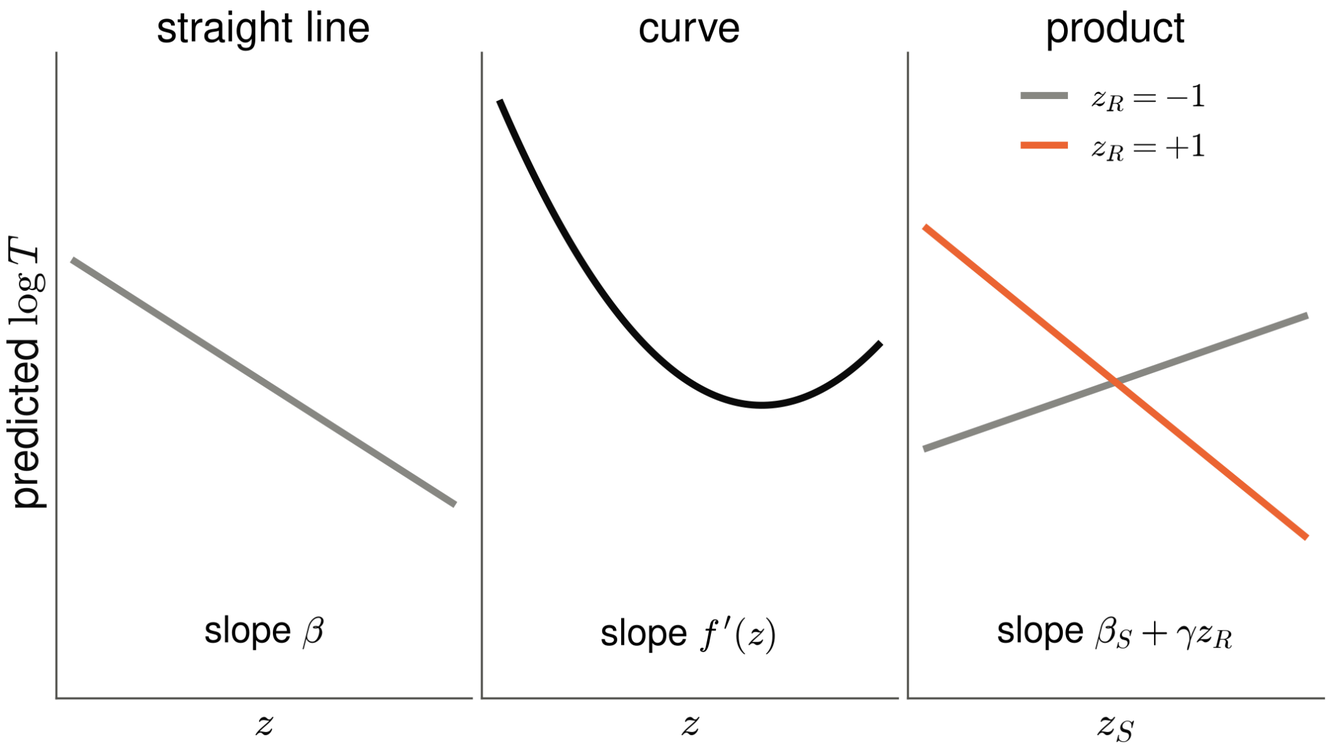
Every model reads the same three statistics at step 2,430. Each one is rescaled to mean 0 and spread 1, giving \(z_A\), \(z_S\) and \(z_R\).

Model	\(\log T \approx\)
Straight line	\(\beta_0 + \sum_k \beta_k z_k\)
Square of \(S_t\)	line \(+\ \beta_{SS} z_S^2\)
Curve for each	\(\beta_0 + \sum_k f_k(z_k)\)
Curves + products	curves \(+ \sum_{j<k} \gamma_{jk} z_j z_k\)
Kernel ridge	\(\sum_i a_i\, e^{-\gamma \lVert z - z_i \rVert^2}\)
Boosted trees	a sum of small decision trees

FOR THE CLOCK

Letting scale bend helps the most. Products and flexible models add little.

A curve changes the slope, and a product links two statistics



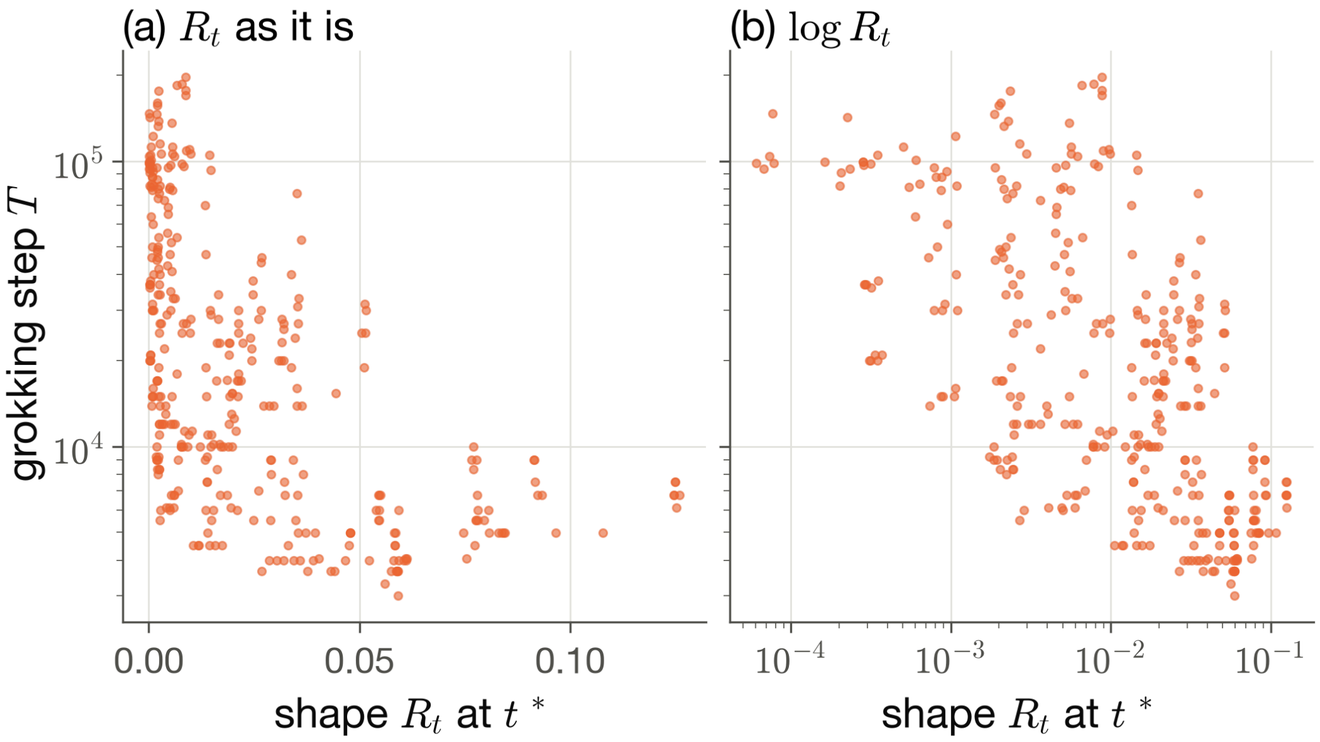
Curve. The slope can change along the way. We replace \(\beta z\) with a cubic spline with 5 knots: \[f(z) = \sum_{j=1}^{7} c_j B_j(z)\] Each \(B_j\) is a fixed cubic bump. The square \(\beta z + \beta' z^2\) is the simplest curve.

Product. How much scale matters depends on shape. We add \(\gamma\, z_S z_R\), so the slope in \(z_S\) becomes \(\beta_S + \gamma z_R\).

FOR THE CLOCK

Best simple formula, \(R^2\) 0.50: \(\log T = \beta_0 + \beta_A z_A + \beta_S z_S + \beta_{SS} z_S^2 + \beta_R z_R\)

Shape enters on a log scale, because \(R_t\) bunches near zero



At \(t^*\), \(R_t\) runs from \(6\times10^{-5}\) to 0.13. 46% of the runs that grok have \(R_t\) below 0.01.

Held-out \(R^2\) of \(\log T\)	**\(R_t\)**	**\(\log R_t\)**
Straight line in shape alone	0.29	0.37
Kernel ridge on shape alone	0.34	0.35
Kernel ridge on all three	0.47	0.55

> On a log scale the runs spread out and \(T\) falls close to a straight line. The log matters most when the statistics are combined.

The runs and the held-out cells are those of Tier 3. A flexible model can bend to follow \(R_t\) alone, so the gap is small there.

Tier 3 on an unseen \(\alpha\) in numbers

Model on a held-out \(\alpha\)	Concordance	Range over seeds
\(A_t\) alone	0.52	
\(S_t\) alone	0.54	
\(\log R_t\) alone	0.55	
Trees on all three	0.70	0.69 to 0.70
Recurrent network on their history	0.76	0.75 to 0.78
Trees on \(\log N\) alone	0.67	0.67 to 0.67
Trees on \(\eta\lambda\) alone	0.72	0.71 to 0.73
Trees on \(N\) and \(\eta\lambda\)	0.79	0.77 to 0.80

Means over the three held-out values of \(\alpha\). The trees have 5 seeds and the recurrent network 3. Each statistic alone enters a linear accelerated failure time model, which has no seed.

How the prediction models were scored

- The statistics \(S_t\), \(\log R_t\) and \(A_t\) are read at \(t^* = 2{,}430\), before any run reaches the grokking level.
- The seeds of a cell are close copies, so whole cells or whole values of \(\alpha\) are held out.
- On held-out cells, kernel ridge regression is fitted to each subset of the statistics. On a held-out \(\alpha\), a linear accelerated failure time model takes each statistic alone, gradient-boosted trees take all three (Chen and Guestrin 2016), and a gated recurrent unit takes their history (Cho et al. 2014).
- The weight decay is a setting, so it is not a predictor. Trees on \(N\) and \(\eta\lambda\), together and each alone, serve as a reference.
- The scores are \(R^2\) and the concordance index (Harrell et al. 1982). Censored runs enter through the likelihood of the accelerated failure time model (Kalbfleisch and Prentice 2002).

Our baseline beside Kumar et al.

- We use their task and settings, which are width 100, the squared error, full-batch gradient descent with $\eta_0 = 100$, $p = 23$ and 90% of the pairs for training (Kumar et al. 2024, App. 8.3, arXiv v3).
- They do not state the parameterisation or the initialisation. We use the NTK parameterisation.
- Our grokking steps at $\alpha = 0.5$ and $\alpha = 1$ are close to those in their Fig. 2. Our networks fit the training pairs several times sooner.
- Every seed groks in 35 of the 48 cells at $\alpha = 0.5$, 26 at $\alpha = 1$ and 8 at $\alpha = 2$.

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